

NEHAL SHAILESH SOLANKI

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PROFESSIONAL SUMMARY

Computer Science student specializing in **backend and full-stack software engineering**, with hands-on experience building **REST APIs, microservices, distributed systems, automated testing, and database-driven applications**. Proficient in **Java, Python, Spring Boot, NestJS, PostgreSQL, MongoDB, Kafka, Redis, Docker, and CI/CD**, with experience implementing **scalable service architectures, and end-to-end testing**. Strong foundation in **Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, and cloud computing**, with additional experience in **AI/ML, RAG, LLMs, and predictive modeling**.

EDUCATION

B.Tech in Computer Science & Engineering (Bioinformatics)

Aug 2023 - Jul 2027

Vellore Institute of Technology (VIT), Vellore

CGPA: 8.72/10

EXPERIENCE

Machine Learning Intern

May 2026 - Jun 2026

Brilliant Software Solutions LLP, Pune

- Developed modular Python-based ML pipelines for data preprocessing, feature engineering, model training, validation, and evaluation across datasets containing **12,000+ customer records**.
- Built reusable Scikit-learn workflows using **Random Forest and XGBoost** with GridSearchCV for hyperparameter optimization, improving prediction accuracy from **86.4% to 91.2%**.
- Evaluated predictive models using **Accuracy, F1-score, and ROC-AUC**, and applied structured problem-solving practices including **testing, data validation, modular pipeline design, debugging & Git-based collaboration**.

AI & Cloud Technologies Intern

Jul 2025 - Aug 2025

IBM SkillsBuild Internship by Edunet Foundation, Remote

- Engineered 2 cloud-based AI applications & **Agentic AI workflows** using **Python, Large Language Models, LangChain, and IBM Cloud**, integrating AI capabilities into data-driven application workflows.
- Established **RAG pipelines** involving document ingestion, data preprocessing, embeddings, vector search, semantic retrieval, and prompt engineering for context-aware responses, gaining practical exposure to **Generative AI**.

PROJECTS

Ready to Revise - AI-Powered Intelligent Revision System

Kotlin | TypeScript | NestJS | Node.js | Prisma |

PostgreSQL | Python | FastAPI | Scikit-learn | Tailwind CSS | Pytest | Docker | CI/CD

- Designed a **microservices-based backend** integrating **5+ behavioral signals**, including student performance, confidence, response time, hesitation, and historical revision behavior, to personalize revision schedules.
- Assembled an ML pipeline for **feature engineering, model training, validation, and prediction**; compared and tuned **Logistic Regression, Random Forest, and Histogram Gradient Boosting**, achieving **0.84 ROC-AUC**.
- Programmed and integrated **15+ REST API endpoints** between the backend and Android application for question management, revision tracking, feedback collection, and **automated scheduling**.

Society Maintenance & Management Platform

Next.js | React | Spring Boot | MongoDB | JUnit | Redis | Kafka

- Architected a full-stack society management platform** with a **56 endpoint REST API**, MongoDB persistence, and role-specific resident and staff workflows.
- Built automated API and end-to-end test coverage** with **37 tests and 66+ end-to-end checks** covering authentication, authorization, CRUD workflows, core business logic, and security scenarios.
- Implemented 4-role authorization and JWT authentication**, complaint tracking, SLA-based overdue detection, and audit logging, validating role-specific access and workflow behavior.

Product Analytics Dashboard

Excel | PostgreSQL | SQL | Power BI

- Designed and optimized **8 analytical SQL views** using joins, aggregations, CTEs, and window functions to analyze sales performance, product ratings, discount trends, seller performance, inventory levels, and return policies.
- Performed **data cleansing, transformation, normalization, and validation** on data using Excel and PostgreSQL.
- Built interactive **Power BI dashboards** with KPI cards, slicers, drill-throughs, and dynamic visualizations.

SKILLS

Languages: Java, Python, TypeScript, JavaScript, Kotlin, SQL

Backend & Distributed System: Spring Boot, FastAPI, NestJS, Node.js, Express.js, Microservices, Kafka, Redis

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

Data, Databases & ORM: PowerBI, Microsoft 365, Excel, PostgreSQL, MongoDB, Prisma, SQL

Cloud, DevOps & Tools: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Git, GitLab, Maven, Terraform

Software Engineering: JUnit, Pytest, Automated Testing, End-to-End Testing, Agile, Scrum

AI/ML: Scikit-learn, Random Forest, RAG, LangChain, LLMs, Agentic AI, Prompt Engineering, Vector Search, GenAI

ACHIEVEMENTS

- Oracle Cloud Infrastructure Generative AI Professional Certification

July 2025 - July 2027

- Events Head, IEEE ITS: Demonstrated **leadership, cross-functional collaboration, & stakeholder co-ordination**.